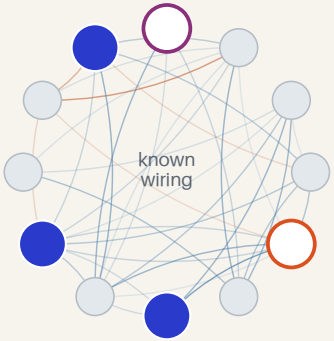
# Refined risk estimates often leave the recording decision unchanged

Choosing 3 neurons to record in order to predict a stimulus response; simulated 12-neuron networks with known wiring



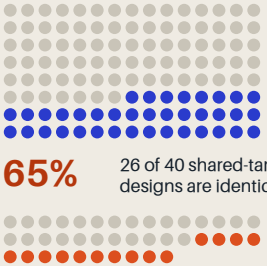

recorded node (budget M = 3)
stimulated node
response readout

Example: network J5000 (1 of 40)
blue = excitatory, orange = inhibitory

## 1 Same recordings?

Do the two criteria choose different data? (40 networks)

**68%** 82 of 120 single-target designs are identical

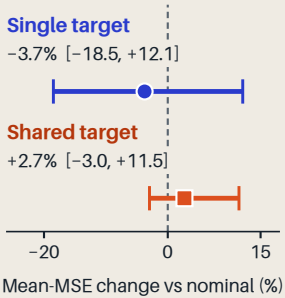


**65%** 26 of 40 shared-target designs are identical

same recording set

## 2 Lower error?

Actual response MSE after fitting (40 networks)

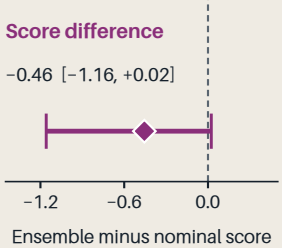



97.5% paired network-cluster CI

## 3 Better forecast?

Same data and prediction (40 new networks)

**19** of 40 networks better, 21 worse




95% paired network-cluster CI

**WHAT THIS SHOWS**

How much prior averaging can help is capped by how often it changes the design. All three primary intervals include zero, so no stable gain is shown. The same three-step audit fits any criterion.

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Simulation only; no biological validation